Global Facilities - Design & Construction Standard - Mechanical - AT Cleanroom HVAC System

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new Cleanroom HVAC Systems for Assembly and Test Sites as part of new Greenfield FAB Construction projects.

1.2 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 The document covers Cleanroom HVAC Systems for Front End Sites including airflow supply, conditioning, and recirculation for control of Cleanroom particles, temperature, humidity, and pressurization. It does not cover other Cleanroom Systems such as Civil, Architectural, Structural (CSA), Fire Suppression, Electrical, Controls, or other Cleanroom Utilities.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron AT sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
AMC	Airborne Molecular Contaminants
ASSY	Assembly
AT	Assembly and Test
CAT	Component Assembly and Test
CHW	Chilled Water
CD	Condensate Drain
DB	Dry Bulb
DCC	Dry Cooling Coil
DP	Dewpoint
FAT	Factory Acceptance Test
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
FFU	Fan Filter Unit
FMCS/SCADA	Facility Monitoring and Control System / Supervisory Control and Data Acquisition
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
HEPA	High Efficiency Particulate Arrest (Filter)
ICA	Instrument Control Air
IST	Integrated System Test
ITP	Inspection and Test Plan
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
MAU	Makeup Air Unit
N+1+F	System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +1 or more future units
OAT	Operational Acceptance Test
PLC	Programmable Logic Controller
POC/LPOC	Point of Connection / Lateral Point of Connection
RA	Return Air

Term/Acronym	Definition / Description
RAC	Return Air Chase
RCU	Re - Circulating Unit
RH	Relative Humidity
SA	Supply Air
SAT	Site Acceptance Test
TUM	Tool Utility Matrix
ULPA	Ultra Low Penetration Air (Filter)
UPS	Uninterruptible Power Supply
VSD/VFD	Variable Speed Drive / Variable Frequency Drive
WB	Wet Bulb

4.0 Roles and Responsibilities

Roles	Responsibilities
Global Facilities Engineering Team - Global (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
Global Facilities Engineering Team - Regional (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to Global Facilities Engineering Team, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

5.1 The following systems are used to support the Cleanroom HVAC:

5.1.1 **Cleanroom Recirculation Air** is the main Cleanroom air that is recirculated to provide control of Cleanroom dry bulb temperature, particle cleanliness levels, and AMC levels. This air is conditioned via:

a. the Dry Cooling Coils, filtered via the FFU ULPA/HEPA filters, and supplied via the FFU fans to the Cleanroom.

b. RCU/AHU with HEPA filters, ducted supply and ducted return to the Cleanroom.

The recirculation airflow in Cleanrooms is used to meet the cooling loads and also to meet the dilution requirement to reduce room particle concentration.

5.1.2 **Cleanroom Makeup Air** is 100% outside air that is supplied to provide control of Cleanroom humidity and pressurization levels. The air is conditioned, filtered, and supplied to the Cleanroom via the Makeup Air Units.

5.1.3 **Cleanroom Exhaust** is air that is removed from the Cleanroom for process/tool exhaust or general space ventilation.

5.2 The main Cleanroom HVAC equipment and components shall be as follows:

5.2.1 Fan Filter Units (FFU)

5.2.2 Dry Cooling Coils (DCC)

5.2.3 Air Handling Units (AHU) / Re-Circulating Unit (RCU)

5.2.4 Return Air Chase

5.2.5 Return Air Plenum

5.2.6 Raised Floor System

5.3 Other Cleanroom Systems associated with Cleanroom HVAC include:

5.2.1 Makeup Air (MAU)

5.2.2 Exhaust

6.0 Cleanroom Configuration

The following shall be the default Cleanroom Recirculation Air design concept for typical AT FAB of CAT CR1, CR2 or whichever module/process required higher cleanroom classification.

For buildings with multiple level AT FABs, the same concept shall be applied to the multiple levels of FAB Cleanroom levels.

The number of levels of a FAB building will be project dependent, but the Cleanroom Recirculation Air design concept shall be the same as outlined in this Standard.

7.0 Cleanroom Specification

7.1 Cleanroom Classification and Particles

The Cleanroom HVAC System shall be designed and operated to meet the Cleanroom Classification Specifications listed in Table 1. This Classification based on particle size measured at 0.3µm. The Cleanroom Classification shall be initially certified at As-Built condition, but the Classification shall also apply through the entire life of the Cleanroom during Operational conditions.

Table 1: FAB Cleanroom and Auxiliary Cleanroom Space Specification - Particles

Parameter	Specification Requirement	Applicable Area [a] / Remarks
Cleanroom Particles	ISO 14644-1 Class 5 (Class 100)	Post-fab Wafer Finish [b] Area
Cleanroom Particles	ISO 14644-1 Class 6 (Class 1k)	CAT Assembly (CR1 & CR2)
Cleanroom Particles	ISO 14644-1 Class 7 (Class 10k)	CAT Assembly (CR3, Post Encap, etc.)
Cleanroom Particles	ISO 14644-1 Class 8 (Class 100k)	CAT: Test, Post-E, etc. MOD/SSD: Assy, Test, etc.

[a] Applicable areas are for reference only, the actual cleanroom classification shall subject to production user specified requirement and refer to "Global AT - Cleanroom Network Protocol Documentation (EN)"

[b] PWF specification will be captured in separate design standard.

7.2 Cleanroom Temperature and Humidity

The AT Cleanroom HVAC Systems shall be designed and operated to meet the Temperature and Humidity Specification listed in **Table 2**.

Table 2: AT FAB Cleanroom Specification – Temperature & Humidity

Parameter	**Specification Requirement**	**Applicable Spaces**
AT FAB Cleanroom		
Cleanroom Temperature	22 ± 3 °C [71.6 ± 5.4 °F]	FAB Cleanroom
Cleanroom Humidity	Design Capability: 50 ± 10 %RH	FAB Cleanroom

Table 3 is a reference table of equivalent Dewpoint Temperatures, based on the Dry Bulb Temperature and Relative Humidity:

- Based on the Design Cleanroom Conditions of 22.0°C DB and 50% RH, the equivalent Dewpoint Temperature is 11.1°C DP; This applies to the AT FAB cleanroom design capability requirement.

Table 3: AT FAB Cleanroom Environment - Equivalent Dewpoint Temperature (for Reference)

Dry Bulb Temp	**Relative Humidity (% RH)**				
	40%	**45%**	**50%**	**55%**	**60%**
19.0°C [66.2°F]	5.1	6.8	8.3	9.7	11.1
20.0°C [68°F]	6.0	7.7	9.3	10.7	12.0
21.0°C [69.8°F]	6.9	8.6	10.2	11.6	12.9
22.0°C [71.6°F]	7.8	9.5	11.1	12.5	13.9
23.0°C [73.4°F]	8.7	10.4	12.0	13.5	14.8
24.0°C [75.2°F]	9.6	11.3	12.9	14.4	15.7
25.0°C [77°F]	10.4	12.2	13.8	15.3	16.7

7.3 Cleanroom Pressurization

The Cleanroom HVAC System shall be designed and operated to maintain the Pressurization Specification listed in **Table 4**.

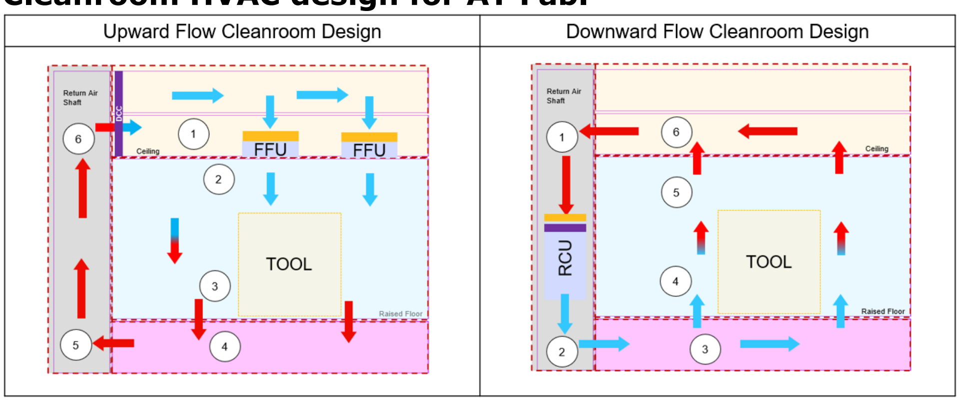
Table 4: FAB Cleanroom Specification - Pressurization

Parameter	**Design Requirement**	**Applicable Sites / Remarks**
Cleanroom Pressurization	Minimum: 7-8 Pa Recommended: 10-12 Pa	FAB Cleanroom **Note:** Pressurization is with reference to external (outdoor) environment.

8.0 Cleanroom Design:

8.1 The AT Cleanroom HVAC system within this document covered 2 types of AT cleanroom HVAC system, which are **Downward Flow Type** and **Upward Flow Type**. Selection of HVAC design are subject to application, served compartment heat load, cleanroom classification, available space, height limitation, etc. In some locations and/or areas, more than one system type maybe an acceptable option. Refer to **Figure 1** for illustration of downward flow and upward flow design.

Figure 1: Illustration of Downward flow and Upward flow Cleanroom HVAC design for AT Fab.



8.1.1 Downward Flow Cleanroom Design

- The downward flow cleanroom design uses a ceiling filter that covers majority of the ceiling surface of the cleanroom. The air flows downward from the fan filter unit towards the floor, where it is recirculated through raised floor and return air shaft.
- This design have better particle control, since the air flow is directed downwards to prevent particles and contaminants airborne from cleanroom floor to production tools.

- This design typically use for higher cleanliness/classification of cleanroom due to higher particle requirement
- However, this design requires a higher ceiling height to accommodate the ceiling filter, dry cooling coil, and the plenum space above for utilities.

8.1.2 Upward Flow Cleanroom Design

- The upward flow cleanroom design uses a floor mounted air handling unit (AHU) and installed with filters to filtered the air before discharge to cleanroom. The air flows upward from the raised floor opening towards the ceiling, where it is recirculated through above ceiling and return air duct.
- This design have poorer particle control, since the air flow is directed upwards. Any particles and contaminants resides at the cleanroom floor might airborne due to upwards flow.
- This design typically use for lower cleanliness/classification of cleanroom but high heat load area. The cooling coil are no longer limited by the height and space above cleanroom ceiling.
- However, this design requires a bigger footprint to accommodate the AHU, and return ducting.

8.2 Typical Cleanroom Airflow Recirculation (FFU)

The Recirculation Air Change rate is used to achieve the FAB Cleanliness specification and to achieve the Cleanroom Temperature and Humidity specification. Providing sufficient number of Air Changes is essential to capture and remove the particles generated in the space, and also to provide cooling and remove the heat load from the space.

The Air Change rate in the FAB Cleanroom space is a function of the FAB Cleanroom volume (Cleanroom Area x Ceiling Height), Ceiling FFU Coverage, and FFU air velocity.

The minimum design requirement and recommended design for FFU Coverage and Air Change Rate is shown in **Table 5**:

Table 5: FAB Cleanroom Environment – Cleanroom Class, Minimum FFU Coverage and Air Change Rate

Space	ISO Class	Minimum FFU Coverage[a]	Minimum Air Change Rate[a]
FAB Cleanroom	ISO Class 5	Minimum: >37% Recommended: >41%	Minimum: > 80 ACH Recommended: > 90 ACH

FAB Cleanroom	ISO Class 6	Minimum: >30% Recommended: >35%	Minimum: > 75 ACH Recommended: > 85 ACH
FAB Cleanroom	ISO Class 7	Subject to cooling load requirement in served module or area of production	Subject to cooling load requirement in served module or area of production
FAB Cleanroom	ISO Class 8		

Note: Table 5 is calculated based on a 6.5m Cleanroom Ceiling height. For a given FFU coverage and FFU airflow amount, a lower Cleanroom ceiling height will provide a higher Air Change Rate. However for Cleanrooms with lower ceiling heights, the higher Air Change Rate may be preferred to push the heat generated by tools downwards.

For upwards airflow cleanroom flow design, the Air Change Rate may be lower as the heat generated by tools are tends to flow upwards to ceiling return grille.

[a] The minimum FFU coverage and minimum air change rate may vary and subject to cooling load requirement in served module or area of production.

8.2.1 To determine the Air Change Rate, the FFU airflow amount shall be calculated based at the FFU air velocity of 0.40 m/s (80 FPM). The FFU selection shall have capability of operating at a minimum of 0.45 m/s (90 FPM), and therefore that the FFU speed can be increased in the future to increase the Air Change Rates.

8.3 Cleanroom Airflow Recirculation through Return Air Chase and Interstitial Plenum

The sizing and design of the vertical Return Air Chase and Interstitial Plenum is important to minimize the pressure losses and enable proper Cleanroom Airflow Recirculation.

This Standard covers the airflow design requirements of the Cleanroom Return Air Chase and Interstitial Plenum, but will not cover the design and construction of Return Air Chase and Interstitial Plenum including building construction, performance and load ratings, structural supports, materials, coatings, constructability, etc.

8.3.1 The velocity in Return Air Chase shall be kept below 2.5 m/s (500 FPM). The recommended design velocity is 2.0 m/s (400 FPM). If these prescriptive conditions cannot be fulfilled, relevant engineering calculation is needed to justify that the pressure drop associated with higher velocity is still within acceptable range per case-by-case basis.

8.3.2 The velocity in Interstitial Plenum shall be kept below 2.5 m/s (500 FPM). The recommended design velocity is 2.0 m/s (400 FPM). If these prescriptive conditions cannot be fulfilled, relevant engineering calculation is needed to justify that the pressure drop associated with higher velocity is still within acceptable range per case-by-case basis.

8.3.3 The typical clearance height above the cleanroom ceiling shall be designed at least 2.0 m (6 feet) to provide a sufficient cross-sectional area for Cleanroom air to recirculate and to accommodate for the utility services to support the production tools.

8.3.4 There shall be no air infiltration or leakage from the outside spaces into the Return Air Chase or Interstitial Plenum. This includes, but not limited to, the following:

- Building construction seams, joints, and gaps shall be covered with separation panels and sealed to prevent infiltration and leakage.
- All joints in walls, floors, and ceilings shall be of vaportight construction.
- Sealing and caulking with cleanroom approved sealant shall be used to seal all gaps between panels and gaps in cut-outs.

Note: The Interstitial Plenum is typically at negative pressure with relation to the Cleanroom FAB and outdoor environment. Therefore, the building structural walls and roof provide the Cleanroom containment area. Negative pressure plenums ensure all air goes through the FFUs and filtration system. However, this introduces the risk of contamination because the building structure is used as walls of the plenum and return air chase. The joints of roof and walls must be sealed vaportight to avoid leakage of unconditioned and unfiltered outdoor air from outside into the Interstitial Plenum or Return Air Chase.

8.4 Cleanroom Airflow Recirculation through Raised Floor

Sufficient floor openings in the Cleanroom Raised Floor is to allow proper airflow path and enable proper Cleanroom Airflow Recirculation.

This Standard covers the airflow design requirements of the Cleanroom Raised Floor tile, but will not cover the design, selection, and construction of the Cleanroom Raised Floor System including system type, performance and load ratings, structural supports, materials, coatings, constructability, etc.

8.4.1 It is recommended to have minimum 60-75% of the Cleanroom floor tile as Perforated Panel type or Grating Panel type, to allow proper Cleanroom Airflow to pass through from below raised floor to Fab raised floor level. The floor opening coverage % should be based on the FFU coverage.

8.4.2 As Cleanroom tools are being installed, the perforated floor tiles shall be relocated from under the tools to in-between the tool aisles, to maintain the cleanroom recirculation air flowpath.

8.4.3 When selecting Cleanroom Raised Floor tiles, the following parameters shall be met:

- Allowable pressure drop shall be between 5 to 10 Pa. The recommend pressure drop is less than 5 Pa.
- The typical airflow rating per floor tile size 600mmm x 600mm (24in x 24in) shall be ~ 500 CMH to 860 CMH each.
- Free area opening for Perforated Panel type: 17-22% opening
- Free area opening for Grating Panel type: 50% opening

8.4.4 In areas with high heat loads and in any other areas where possible, it is recommended to use Grating Panel type floor tile, which has 50% free area opening to allow more Cleanroom Airflow to pass through.

8.4.5 The typical clearance height under the raised floor shall be designed between 0.9 to 1.7 m (3 to 5.5 feet) to provide a sufficient cross-sectional area for Cleanroom air to recirculate back into Return Air Chase and to accommodate for the utility services to support the production tools.

8.4.6 The air velocity under the raised floor shall be checked, and relevant engineering calculation is needed to justify that the pressure drop will be within the acceptable range per case-by-case basis.

8.5 Cleanroom Airflow Supply through Raised Floor and Recirculate via Cleanroom Ceiling

Sufficient floor openings in the Cleanroom Raised Floor is important to allow proper airflow path and enable proper Cleanroom Supply Airflow and Recirculation.

This Standard covers the airflow design requirements of the Cleanroom Raised Floor tile, but will not cover the design, selection, and construction of the Cleanroom Raised Floor System including system type, performance and load ratings, structural supports, materials, coatings, constructability, etc.

8.5.1 It is recommended to have minimum 60-75% of the Cleanroom floor tile as Perforated Panel type or Grating Panel type, to allow proper Cleanroom Airflow to pass through from below raised floor to Fab raised floor level. The floor opening coverage % should be based on the air flow requirement for served area.

8.5.2 As Cleanroom tools are being installed, the perforated floor tiles shall be relocated from under the tools to in-between the tool aisles, to maintain the cleanroom recirculation air flowpath.

8.5.3 When selecting Cleanroom Raised Floor tiles, the following parameters shall be met:

- Allowable pressure drop shall be between 5 to 10 Pa. The recommend pressure drop is less than 5 Pa.
- The typical airflow rating per floor tile size 600mmm x 600mm (24in x 24in) shall be ~ 500 CMH to 860 CMH each.
- Free area opening for Perforated Panel type: 17-22% opening
- Free area opening for Grating Panel type: 50% opening

8.5.4 In areas with high heat loads and in any other areas where possible, it is recommended to use Grating Panel type floor tile, which has 50% free area opening to allow more Cleanroom Airflow to pass through.

8.5.5 The typical clearance height under the raised floor shall be designed between 0.9 to 1.7 m (3 to 5.5 feet) to provide a sufficient cross-sectional area for Cleanroom air to supply from AHU via perforated tiles without major obstruction and to accommodate for the utility services to support the production tools.

8.5.6 The air velocity under the raised floor shall be checked, and relevant engineering calculation is needed to assess the risk of airborne particles from raised floor and to justify that the pressure drop will be within the acceptable range per case-by-case basis.

8.6 Cleanroom Heat Load

The Cleanroom HVAC System shall be designed to provide sufficient cooling and airflow provisions, to satisfy the Cleanroom Heat Load and maintain the Cleanroom Dry Bulb Temperature within specifications.

8.6.1 The default Cleanroom Design Heat Load shall be 1,000 W/m2, as shown in **Table 6.**

Table 6: Design Heat Load for Cleanroom Spaces

Parameter	**Minimum Design Requirement**	**Applicable Spaces / Remarks**
Cleanroom Design Heat Load	Recommended: 1,000 W/m2	AT Cleanroom
Cleanroom Design Heat Load	Recommended: 1,200 W/m2	BI Oven Area
Cleanroom Design Heat Load	Recommended: 3,000 W/m2	SLT Area Recommended upwards flow design.

Note: For SLT area calculated heat load is exceed 3,000 W/m2, liquid cooled for production tools shall be considered.

8.6.2 During the Project Design Phase, the actual heat load shall be calculated and verified based on the actual Space layout (lighting, FFUs, etc.) tool layout and tool types, and heat output of all tools and equipment.

Note: For Cleanroom spaces with higher heat load based on the actual tool layout and density, the specified Cleanroom design Heat Load shall be higher to meet the actual heat load requirements.

If these Cleanroom spaces also require a higher Air Change Rate based on the actual tool layout and density, the specified design Cleanroom Air Change Rate shall be higher to meet the actual heat load requirements.

For Cleanroom spaces with lower heat load based on actual tool layout and density, the specified Cleanroom Design Heat Load of 1,000 W/m2 shall be used. This allows for sufficient cooling capacity for future toolsets and scenarios.

8.6.3 Certain equipment inside the cleanroom may generate high amounts of heat and can cause localized temperature fluctuations as a result of effects of radiant heat. It is recommended to locate the Cleanroom Temperature and Humidity sensors away from these tools.

8.7 Cleanroom Pressurization

To prevent or minimize particle migration into the FAB Cleanroom from surrounding areas, the FAB Cleanroom shall be kept positively pressurized relative to the surrounding areas. This positive pressure differential shall ensure that air leakage and particle migration will occur outwards between the Cleanroom enclosure and the dirtier surrounding areas.

8.7.1 The FAB Cleanroom Pressure shall be a minimum of 7-8 Pa. The recommended Cleanroom Pressure is 10-12 Pa. Refer to **Table 4** above.

Note: Traditionally, a pressure differential of 12.5 Pa (0.05" w.c.) has been used for Cleanroom design. Industry reference recommends a minimum space pressure of 10 Pa, to minimize particle migration that may potentially occur from opening and closing of Cleanroom doors. However, for Cleanroom operation with door-closed conditions, industry reference recommends a minimum space pressure of 5 Pa. It states that 5pa pressure levels have proven to be sufficient to minimize the particle migration, and higher values of space pressurization will not result in any extra reduction of Cleanroom particle levels.

Some Micron FABs have been able to operate at Cleanroom pressure levels of ~5 to 6 Pa, with no known issues or problems to meet Cleanroom Classification levels, particle count limits, or wafer particle defects.

8.7.2 This FAB Cleanroom Pressure levels shall be measured relative to outside ambient pressure, via the Zero Reference Pipe (ZRP).

Differential pressure sensors shall be used to measure the difference between Cleanroom pressure and outdoor air, via the ZRP. The ZRP provides a single common "zero" point that all areas can reference.

The ZRP piping shall be leaktight stainless steel piping. The ZRP pressure element shall be installed in a stable location outdoors, and protected from potential effects from weather elements including wind, rain, etc.

8.8 Cleanroom Dry Cooling Coils

The Cleanroom Dry Cooling Coils (DCC) provide cooling for Cleanroom Temperature control. The Cleanroom Dry Cooling Coils shall be in vertical configuration, between the Return Air Chase and Interstitial Plenum Return to the Ceiling FFUs.

8.8.1 The Cleanroom Dry Coil shall be sized, selected, and specified in accordance with **Table 7**.

8.8.2 Medium temperature Chilled Water (MTCHW) shall be provided to the DCC, at supply temperature 12°C or 13°C. The Chilled Water supply temperature shall always be higher than the Cleanroom Dewpoint temperature, to prevent condensation from the piping or coils.

8.8.3 Temperature transmitters shall be provided and installed at the discharge side of each set of Dry Cooling Coils, to monitor the DCC leaving air temperature as it is supplied to Cleanroom.

8.8.4 VESDA smoke detection shall be provided and installed at the return side upstream of the Dry Cooling Coils, for early smoke detection.

8.8.5 All space around the Cleanroom Dry Coils shall be blanked off and sealed, to ensure that all cleanroom recirculation airflow will route through the Dry Coils. There shall be no bypass air around the Dry Coils that does not pass through the coils.

8.8.6 Piping connections to each Dry Coil unit shall have pressure gauges on supply and return, temperature gauges on supply and return, flexible connections at supply and return, and strainer on the supply connection.

8.8.7 Dry Cooling Coils installed above the Cleanroom Ceiling directly in the FAB Cleanroom Space, or in the horizontal plane of the Cleanroom Ceiling in the FAB Cleanroom Space shall be minimized. This configuration would create a risk of possible water leakage or condensation, which would result in interruption or damage to Cleanroom tools, equipment, and AMHS.

8.8.8 The water piping installed directly above the cleanroom ceiling shall be minimized, as much as possible. If unavoidable, dull welded piping joint shall be used.

Table7 : Dry Cooling Coil Specification for Sizing and Selection

Dry Cooling Coil Specification
Cleanroom Dry Cooling Coil (DCC) shall be selected and specified as follows: 1. The Cleanroom Dry Coil shall be selected based on the following parameters: - Entering Air Temperature: 24.0°C DB / 15.3°C WB (9.2°C DP) - Leaving Air Temperature: 16.0°C DB - 20.0°C DB - Entering Water Temperature: 12 or 13°C - Leaving Water Temperature: 19 or 20°C 2. The velocity in Cleanroom Dry Coil shall be kept below the maximum of 2.5 m/s (500 FPM). The recommended design velocity is 2.0 m/s (400 FPM). 3. The pressure drop through the Cleanroom Dry Coil shall be kept below the maximum of 50 Pa (0.20” w.g.). The recommended pressure drop is less than 40 Pa (0.16” w.g.) 4. Cooling coil sizing and capacity shall be determined according to the required air volume and cooling capacity. It is recommended to use coils with minimum 2-3 rows to achieve the required cooling capacity. 5. Coil shall be drainable with threaded plugs at the headers to allow venting and draining. Provide the drains and vent taps. 6. Coil header shall have inlet connections at the bottom of supply headers at the “air-off” or “air-on” face of coils (as specified) and outlet connections at the top of return headers at the “air-on” face of coils. 7. Air Leakage: a. Minimize air leakage through gaps between the fin ends and frames. b. Use baffle plates to close the space between the coil frame and the surrounding structure or equipment. c. Seal gaps between coils and surrounding structure or equipment using non-hardening mastic.

Dry Cooling Coil Specification
8. Drain Pans: a. Provide stainless steel drain pans with drain connection for cooling coils. Provide intermediate drain pans for stacked banks of cooling coils. b. Extend drain pan 3 inches from face of coil entering air side, and 6 inches from face of coil leaving air side, and 4 inches from face of eliminators. c. Install drain piping from the drain pans to collect and remove any condensate, water, etc. 9. Pre-completion Production Tests: a. Factory shall perform Pneumatic leak test; Pneumatically test coils for leakage by submerging in warm water and applying the test pressure, or by static water pressure for at least 1 hour. Certification test certificates from coil manufacturer required for each coil. b. Perform full system test at minimum test pressure of 10 Bar. 10. Installation a. During delivery, assembly, and installation, protect coils so fins and flanges are not damaged. Replace loose and damaged fins. Comb out bent fins. b. Install coils so that fluid and air flow directions are counter flow. c. Support coils using prefabricated aluminum alloy or stainless steel frames **Acceptable Manufacturers –** Aerofin, Johnson Controls, Huntair, Heatcraft, Trane, Precision Coils, or approved equal

8.9 Cleanroom Fan Filter Units

Fan Filter Units (FFU) are used to provide final filtration of Cleanroom Recirculation air and then to supply the air to the Cleanroom FAB Level. FFUs shall draw in air from the above cleanroom ceiling, interstitial plenum and return air shaft, filter the air through the HEPA filters, and then supply air into cleanroom at constant airflow rate.

Fan Filter Units provide high reliability, low first cost, and low operating cost. If one FFU fails, only a very small percentage of the overall air volume to the Cleanroom will be lost. The failed unit can then be easily identified by the FFU Control system for troubleshooting and replacement

8.9.1 The Fan Filter Unit shall be sized, selected, and specified in accordance with **Table 8**.

8.9.2 For the Cleanroom Air Change Rate calculation purposes, the speed of 0.40 m/s (80 FPM) shall be used.

8.9.3 The FFU shall be capable to accommodate a future frame above the FFU air inlet opening, to add an AMC filter.

8.9.4 The FFUs shall be arranged across the Cleanroom ceiling to provide uniform airflow throughout the entire Cleanroom. For example, 50% FFU coverage would be arranged in a "diamond X-grid" pattern so that every other ceiling tile contains an FFU.

8.9.5 During initial construction and initial start-up, a temporary blanket type construction pre-filter shall be used to cover the FFU inlet air opening. This will protect and lengthen the lifetime of the FFU HEPA filter. After construction is complete, the temporary pre-filter shall be removed.

Table 8: Fan Filter Unit Specification for Sizing and Selection

Fan Filter Unit Specification
The Fan Filter Unit size shall match the Cleanroom Ceiling size. Nominal FFU size is 1200mm x 1200mm (48" x 48"). Fan Filter Unit (FFU) shall be selected and specified as follows: 1. Provide fully factory-assembled, packaged, pre-wired FFU units consisting of: a) intake opening b) direct-drive fan with high performance impeller, backwards curved blades, dynamically balanced c) DC Brushless motor (also known as Electronically Communicated (EC) Motor)

Fan Filter Unit Specification
d) addressable controller with feedback e) control software f) sound attenuation liner g) ULPA filter 2. The Fan Filter Units shall be selected with minimum operation range capability of 0.20 m/s to 0.45 m/s (90 FPM). It is recommended to select with operation capability of up to 0.55 m/s for higher performance. 3. The HEPA filter shall be rated H14, 99.995% efficient for Most Penetrating Particle Size (MPPS) of 0.2 microns size. 4. The initial pressure drop rating for HEPA Filter shall be less than 120 to 170 Pa @ 0.45 m/s. It is recommended to be less than 130 Pa @ 0.45 m/s. 5. The FFU shall have minimum fan static pressure rating of 200 Pa. The recommended fan static pressure rating is greater than 250 Pa. 6. FFU shall be compatible with the Cleanroom ceiling grid system specified. FFU shall fit into ceiling grid geometry without any gaps or leakage. 7. Units shall be designed with sufficient structural strength to be capable of being supported at each corner. Units shall be walkable and support loading on top of unit. 8. FFU shall be manufactured in a clean environment 9. Fan: a) Fan shall be high efficiency centrifugal plug fan with EC motor, variable flow capacity, direct drive with backwards curved centrifugal impeller, sealed bearings, external rotor motor rated for continuous duty, furnished with internally wired thermal overload protection. Bearing lifetime shall be 80,000 hours. 10. FFU Control System shall be provided with fully integrated software package, tested in service, for monitoring and control of FFU operation, which includes: 1. Unit starting and stopping 2. Adjusting speed set point 3. FFU status monitoring and alarms 4. FFU runtime counter a) In the event of FFU remote monitoring network or power failure, configure controls so unit operates at last condition

Fan Filter Unit Specification
b) Each FFU shall be able to operate individually by a central control and monitoring system c) FFU shall have indicator lights to indicate status of malfunction. 11.Motors: a) Provide high efficiency electronically commutative (EC) brushless DC type motors with built-in power factor controller. b) Provide motors of protection class IP44, motor insulation class A. 12. Filter: a) Provide High-Efficiency Particulate Air (HEPA) filter, with minimum DOP efficiency of 99.995% rated on 0.2 microns. b) HEPA Filter material shall be microfiber glass, or PTFE media. c) The filter media preferable to be FM approved. d) Provide FFU with gasket seal between the filter and the FFU housing e) Filters shall be factory tested to demonstrate compliance with efficiency, leak, and pressure drop specifications. Provide all filters with factory test certificate of compliance 13. Provide FFU with the following Labels and Listings: a) UL 507 / CAN CSA C22.2 no. 113-M 1984 b) UL 2111 c) UL 1917 / CAN CSA C22.2 d) All electrical components shall be UL listed and labeled. **FFU Approved Manufacturers:** M+W Products (Exyte), Nortek Air Solutions, Gebhardt, AAF Flanders, or approved equal **HEPA Filter Acceptable Manufacturers:** AAF, Camfil, Trox, or approved equal

8.10 Cleanroom Makeup Air

Makeup air is used to provide Cleanroom Humidity (i.e. Dewpoint) control as well as maintain positive pressurization inside the Cleanroom to prevent particle contamination and air infiltration. Makeup air shall be provided to compensate for the exhaust and cleanroom leakage.

This Standard covers the Makeup Airflow design requirements corresponding to proper Cleanroom Design. However, this Standard will not cover the design, selection, and construction of the Cleanroom Makeup Air Units including equipment and system type, performance ratings, materials, coatings, constructability, etc. Refer to the Cleanroom Makeup Air Design and Construction Standard for this information.

The minimum Makeup Air provided shall be calculated per **Table 9**:

Table 9: Guideline for Cleanroom Makeup Air Requirement Calculation

- Cleanroom Makeup Air Requirement = Total Exhaust inside Cleanroom + Total Pressurization/ Leakage

- Total Exhaust inside Cleanroom for respective level = Tool Exhaust + Tool Exhaust inside Cleanroom + Space/Area Exhaust inside Cleanroom

- Total for Pressurization/Leakage = 1 Cleanroom Air Change per hour (based on Cleanroom Volume x 1)

8.10.1 Create a Room Pressure and Flow (P&F) diagram for the Cleanroom, to show the following:

a. Airflow design settings (values) of all supply, return, and exhaust amounts
b. Desired room pressure value and tolerance
c. Leakage flow directions due to room pressure differentials

The Cleanroom Air Balance calculation shall be provided to show the amount of Makeup Air required versus Exhaust Air, Cleanroom Leakage, and Pressurization at Full Build conditions

8.10.2 Makeup air shall be ducted and supply to the cleanroom ceiling supply air grille.

8.10.3 Sufficient mixing shall be ensured between makeup air and recirculation air to prevent temperature and humidity stratification and imbalance.

8.10.4 Makeup air supply shall have capability to be provided at following parameters:

- Normal Makeup Air Supply Design Setpoint shall be Dry Bulb 23°C, Dew Point 11.1°C

    - The Makeup Air Supply Dry Bulb Temperature shall have capability to provide as low as 16°C, and capability to provide as high as 23°C.
    - The Makeup Air Supply Dew Point Temperature shall have capability to provide as low as 9.0°C and capability to provide as high as 13.8°C.

    During winter seasons, it may be required to supply the Makeup Air discharge at a higher Dewpoint temperature than the Cleanroom Dewpoint temperature, to provide additional humidification:

    - Dew Point Temperature: Makeup Air Supply shall have capability to provide as high as 13.8°C for humidification.

    During summer seasons, it may be required to supply the Makeup Air discharge at a lower Dewpoint temperature than the Cleanroom Dewpoint Temperature, to provide additional de-humidification:

    - Dew Point Temperature: Makeup Air Supply shall have capability to provide as low as 9.0°C for dehumidification.

8.10.5 For sites with relatively constant climate year-round (i.e. Singapore, Malaysia), either fixed control setpoint or Auto-Cascade control setpoint can be considered for the Makeup Air discharge dewpoint controls.

8.10.6 For sites with seasonal climate changes for summer, winter, fall, and spring, (i.e. USA, Japan, Taiwan, China), it is recommended to use Auto-Cascade control capability to automatically adjust the Makeup Air Discharge Dewpoint setpoint.

a. Upper and Lower limit clamps shall be set to prevent Makeup Air Discharge Dewpoint from drifting excessively from the normal setpoint.

b. Upper and Lower alarm limits shall be set to notify Operations if the Makeup Air Discharge Dewpoint drifts excessively from the normal setpoint.

8.10.7 The Makeup air unit shall have the following filters and filter ratings, at a minimum:

a. Pre-filter rated at 25 to 35% (MERV 7)
b. Intermediate filter rated at 85% (MERV 13)
c. Final filter rated at 99.95% (HEPA H13)
d. To reserve the space for 2 Racks of AMC filters for Ozone, Acids, Bases, as needed based on the site's ambient air conditions and upstream AMC challenge in the future to cater for technology advancement and future process needed.

Note: Makeup air units with air washers will have capability to remove SO2 and NH4 via the air washer.

8.10.8 Makeup air intakes shall be placed a minimum of 7.6m (25 feet) away from exhaust discharge locations from the Cleanroom. The air intakes shall not be placed immediately downstream under prevailing winds of exhaust discharge airstreams.

8.11 Cleanroom CFD Modelling

During Design phase, the Cleanroom conditions shall be checked using Computational Fluid Dynamics (CFD) modelling software. CFD modelling shall be used to simulate and determine the following Cleanroom conditions, and make the necessary improvements to the Design:

a) Airflow pattern
b) Temperature distribution
c) Particle concentration
d) Relative humidity distribution
e) Impact of room pressurization

8.12 Cleanroom Temperature Control Zoning

The Cleanroom shall be separated into temperature control "zones" based on the location of Dry Cooling Coils and their corresponding Return Air Chase. Feedback from the Cleanroom Temperature and Humidity (T&H) sensors shall control a single Dry Coil unit or group of Dry Coil units. Refer to **Figure 2** for example of Cleanroom Temperature Control Zoning.

8.12.1 Approximate Cleanroom Temperature Control Zone size shall be 500 to 750m2 each.

8.12.2 For downward flow cleanroom design, the zone shall be controlled by the Dry Cooling Coils facing that zone.

8.12.3 For upward flow cleanroom design, the zone shall be controlled by the Air Handling Unit served to that zone.

8.12.4 Distance from the Return Air Chase to end of zone shall be limited to below 75m distance. It is recommended to limit the distance to between 50m and 60m.

8.12.5 Each Cleanroom zone shall have 2 T&H sensors. The sensors and their respective probes shall be located in a manner that the temperature readings are similar. This is to provide better temperature control via averaging function.

8.12.6 Avoid stagnant areas throughout the Cleanroom, in which there would be no supply of conditioned air from Dry Cooling Coils. The design and location of Return Air Chases and Dry Cooling Coils shall be configured to provide uniform and even distribution of conditioned supply air throughout the entire Cleanroom.

8.12.7 There shall be one control valve station for each set of Dry Cooling Coils, associated with a Temperature Control Zone. The Control Valve station shall have isolation valves, and a smaller size bypass with isolation valve around the control valve.

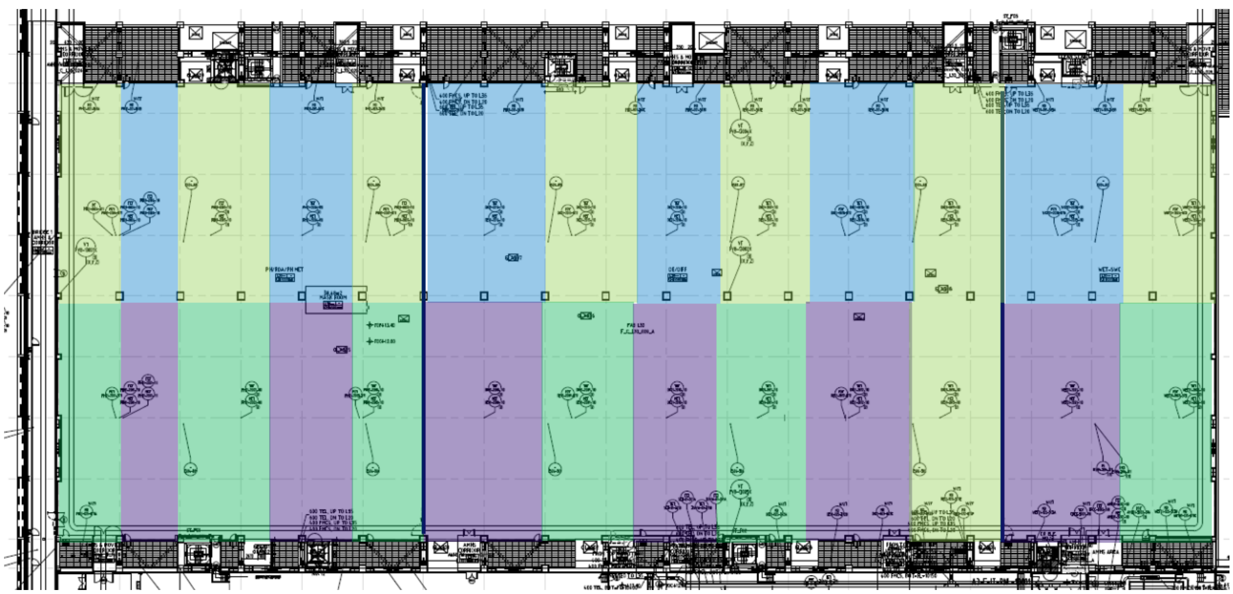


Figure 2 – Example Cleanroom Temperature Control Zoning for FAB Ballroom

8.13 Cleanroom Temperature & Humidity Sensors

Ceiling mounted Temperature and Humidity Sensors shall be used in each Temperature Control Zone.

8.13.1 Downward Flow Cleanroom Design

T&H Sensor shall be located at directly below the FFU discharge, at a height of 100mm below ceiling height, and at a distance 200mm into the face of the FFU. Refer to **Figures 4a and 4b** for example.

This sensor location allows the sensor to sense adequate mixing of the FFU supply air (from the DCC) mixed with the Cleanroom Heat sources.

Regarding temperature sensor location, Industry reference recommends that it is more important to have precise control rather than accurate control, and to provide for repeatability of results. Therefore, the temperature sensors are typically located in plenum area (measuring coil discharge temperature) or just below the Cleanroom ceiling, to measure the temperature of the supply air.

a. Sensor shall be located away from AMHS track.
b. Sensor shall be located away from other obstructions and heat sources, such as exhaust ductwork, utility piping, etc.
c. Transmitter module shall be mounted up at above cleanroom ceiling along catwalk or grating, to allow easy access for calibration and troubleshooting. Refer to **Figure 3** for example of Transmitter mounting detail.
d. The probe shall have excess cable length of minimum 10m, to allow for future relocation of the probe.

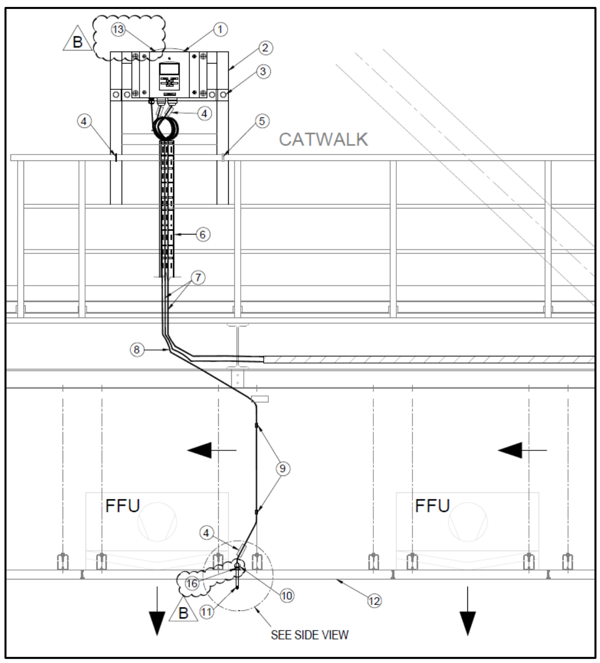
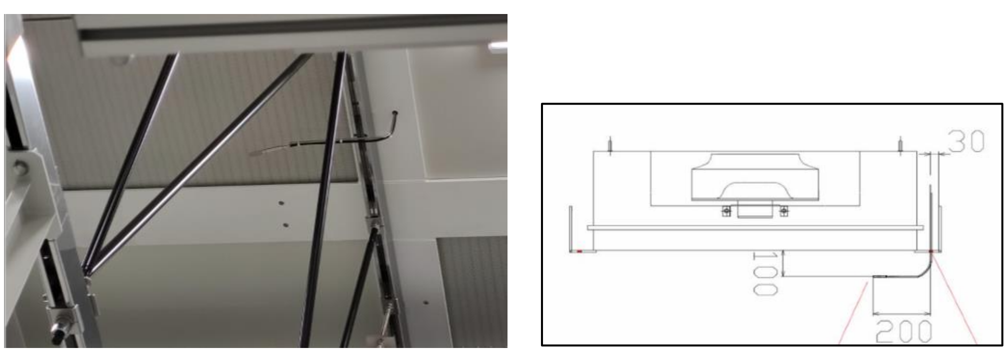


Figure 3 – Example Cleanroom T&H Transmitter and Probe Mounting Detail



Figures 4a & 4b – Recommended Cleanroom T&H Probe Mounting Location

8.13.2 Upward Flow Cleanroom Design

T&H Sensor shall be located at a height of 1.5m mounted on pole per temperature control zone.

This sensor location allows the sensor to sense adequate mixing of the RCU supply air (supply from below raised floor via perforated floor tile) mixed with the Cleanroom Heat sources.

Regarding temperature sensor location, Industry reference recommends that it is more important to have precise control rather than accurate control, and to provide for repeatability of results. Therefore, the temperature sensors are located in 1.5m height to measure the temperature of the supply air.

a. Sensor shall be located away from other obstructions and heat sources, such as production tool heat vent, utility piping, etc.
b. Transmitter module shall be mounted up on pole to allow easy access for calibration and troubleshooting.
c. The probe shall have excess cable length of minimum 10m (excess cable to be coil up and located below raised floor), to allow for future relocation of the probe.

8.14 Cleanroom Building Envelope and Cleanroom Ceiling

This Standard will not cover the design, selection, and construction of the Cleanroom Building Envelope and/or Ceiling System including system type, performance and load ratings, structural supports, materials, coatings, constructability, etc.

The Cleanroom envelope shall be sealed to prevent any infiltration leakage from outside space, or exfiltration leakage to outside space.

It is critical to ensure the Return Air Plenum and Plenum is sealed, as these spaces may be at negative pressure relative to outside. If the spaces are not sealed properly, they may cause outside air to infiltrate and affect Cleanroom Temperature and Humidity control, or load up the FFU HEPA filters with particulate.

Cleanroom ceiling considerations to be provided include, but is not limited to, the following:

a) Ceiling shall have proper gasketing to ensure it is leak-tight and air-tight when pressurized.
b) FFUs shall sit directly in the Cleanroom Ceiling.
c) Cleanroom Ceiling shall have blank walkable access panels in locations without FFUs.
d) Cleanroom Ceiling shall have suitable penetrations for components and devices. These include, but are not limited to,

temperature, humidity, and pressure sensors, sprinkler heads, lighting, I&C, telecom, and other life safety devices.

e) Sheet metal flashing, closures, gaskets and sealing at columns, perimeter walls, and penetrations through blank panels.

f) Changeable layout and locations for filters, blank panels, and luminaries.

g) The space above the Cleanroom ceiling shall have suitable tie-off fall protection, that is OSHA compliant.

For Cleanroom Segregated areas, the different area of the FAB with different cleanroom classification shall be divided into different areas by means of wall separations, floor separations, plenum separations, and ceiling separations. These separations shall allow for dedicated individual temperature, humidity and pressure zoning. Ceiling separator shall be installed from ceiling grid framework up to the ceiling purlins. The plenum separators shall be installed from the ceiling purlins up to the plenum cap. Maintenance Access Panels should be provided to access these enclosed areas.

8.15 AT FAB Air Shower Design

The Cleanroom HVAC System for the FAB Air Shower shall be designed according to section Air Shower Protocol in **Global AT – Cleanroom Network Protocol Documentation (EN)** documents.

8.15.1 Minimum requirement for air shower implementation is all Cleanroom ISO Class 5 and Class 6. (Inbound and outbound)

8.15.2 Air shower direction and angle coverage is recommended to encompass knee level to head level of minimum 6 feet height.

8.15.3 Specifications as shown in **Tables 10** is for reference purposes only. Detail of AT FAB Air Shower Design shall follow the specifications stated in section Air Shower Protocol in **Global AT – Cleanroom Network Protocol Documentation (EN)** documents.

Table 10: AT FAB Air Shower Specifications

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Particles	As per served cleanroom classification	
Cleanroom Temperature		
Cleanroom Humidity		
Air velocity at Nozzle	Minimum 20-25 m/s	
Air Shower Timer	20 sec blow 5 sec stabilization	Air shower door shall come with interlock function. Only unlock after air shower cycle completed.

8.16 Emergency De-Smoke Exhaust System

An Emergency De-smoke exhaust system shall not be required, unless specifically required by local code or AHJ for emergency smoke removal purposes.

If an Emergency De-smoke exhaust system is required, the minimum design flowrate of smoke purge shall be 55m3/h/m2 for cleanroom areas, as outlined in NFPA 318-2002 Edition, Annex A, Clause A.5.6.4, or as required by local code and AHJ.

The Emergency De-smoke exhaust system design and specification requirements will not be covered in this Standard.

8.17 Other General/Miscellaneous:

8.17.1 **Code Requirements -** Cleanroom design shall comply with local code and AHJ requirements based on the level of Hazardous Occupancy Classification.

8.17.2 **Equipment Layout –** All Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment and to allow move-in and move-out of large components during maintenance and repair (such as coils, motors, filters, etc.).

8.17.3 **Water Piping –** Water piping shall be minimized directly above Cleanroom, as much as possible.

8.17.4 **Fire Sprinkler -** FAB Cleanroom shall be covered via Wet Pipe type Fire Sprinkler system. The requirements for whether Fire Protection is required in the Interstitial Plenum depends on the local code and AHJ. Refer to the Fire Suppression Design and Construction Standard for additional requirements.

9.0 Design for Sustainability

The following energy efficiency and Sustainable design elements shall be included:

9.1 **Cleanroom Air Recirculation Low Pressure Drop design –** Design and selection of recirculation shall be to minimize air velocity and pressure drop, to minimize FFU fan energy consumption. This includes lower air velocities through the return air chase, above cleanroom ceiling, interstitial plenum, and across the Dry Cooling Coils. Lower pressure drop FFU HEPA filters. Increasing perforated area openings of raised floor panels, supply and return grille.

9.2 **Minimize Cleanroom Leakage and Pressurization –** The amount of cleanroom leakage shall be kept as minimal as possible via tight building construction. This will allow the Cleanroom pressure levels to be kept lower, and to reduce the amount of makeup air required for pressurization and also to compensate for temperature/humidity impact from infiltration.

9.3 **Minimize Cleanroom Exhaust and Makeup Air –** The amount of Cleanroom exhaust shall be kept as minimal as possible via proper balancing of production tool exhaust and reducing the loads wherever possible. This will allow reducing the amount of makeup air required to compensate for exhaust.

9.4 **FFU Adjustable Speed Controllers –** The FFUs shall be equipped with EC type motors for higher energy efficiency, and adjustable speed controllers to allow more FFU efficient operation at part speed.

10.0 Design for SMART Facilities

The following SMART Facilities design elements shall be considered:

10.1 **Cleanroom Particle Monitoring:** It is recommended to monitor Cleanroom Particle levels via online continuous monitoring system.

11.0 Accessories:

11.1 The following accessories which shall be installed: See **Table 11.**

Table 11: Accessories to be installed

Accessories	Type and Installation Location	Remarks
Piping Insulation	• Install for all CHW piping throughout the system	In accordance with Construction Spec Section 230719 (or equivalent Spec)
Ductwork Insulation	• Install for all MAU Supply Air ductwork in unconditioned spaces or utility area spaces	In accordance with Spec Section 230713 (or equivalent Spec)
Piping Supports	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 230529 (or equivalent Spec)
Flexible Connections	• Install at all CHW connections to Dry Cooling Coils • Install at all ductwork connections to MAU units and equipment	
Pressure Gauges	• Install at all CHWS/R connections to Dry Cooling Coils	
Temperature Gauges	• Install at all CHWS/R connections to Dry Cooling Coils	
Air Vents	• Air vents shall be minimum size DN20 (3/4 inches) • Install Auto Air Vents at all high point locations for CHW	

Accessories	Type and Installation Location	Remarks
	• Install Auto Air Vents at main CHW piping and vertical CHW piping connected to the Dry Cooling Coils	
Drain Ports	• Install at all low point locations • Install at all Dry Cooling Coils	
Valves	• Install valves at all Dry Cooling Coils to allow isolation. • All butterfly valves shall be rated for "bi-directional" duty with positive shutoff in both directions • All butterfly valves shall be lug-style	In accordance with Spec Section 232113 (or equivalent Spec)
Control Valves	• Provide control valve type and quantity as outlined in this standard	
Condensate Drain	• Drain piping shall be provided to collect condensate from the condensate system, and to discharge to the Industrial Waste (IW) system or other approved system.	
Labeling and Identification	• "CHWS" and "CHWR" shall be used for Chilled Water Supply and Return • "MAU" shall be used for Makeup Air Supply Ductwork • Provide identification labels at minimum of every 6m distance and change of direction • Labels for CHWS and CHWR piping shall be Green Background with White Lettering	In accordance with Spec Section 230553 (or equivalent Spec)

12.0 Electrical Design:

12.1 **Power Segregation** – Electrical power supply for Makeup Air Units and FFUs shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.

a) The Electrical power wiring for FFUs shall be arranged in an "alternate" configuration, so that every other FFU is from a different power source. This way, upon the loss of one power source, every other "alternate" FFU would still continue operation to provide uniform airflow across the Cleanroom.

12.2 **Un-interrupted Power for Controls** – 100% of Cleanroom HVAC system controls including PLCs, Local Control Panels (LCPs), FFU System Controller, FFU Control Power, and other auxiliary control power shall be connected to an Uninterrupted Power source (such as UPS or DC Bank) to allow continuous operation of the Cleanroom HVAC system during momentary power interruption or voltage sags, spikes, and fluctuation events.

12.3 **Emergency Power - MAU** – 50% of MAU units shall be on Emergency power and shall continue to run during a normal power loss. This is to maintain positive pressurization in the cleanroom and compensate for the exhaust that will still be operating during a power loss. Refer to the Makeup Air Unit System Standard for additional details and requirements.

12.4 **Emergency Power - FFU** – 50% of FFUs shall be on Emergency power and shall continue to run during a normal power loss. This is to maintain minimum cleanroom recirculation airflow and cleanroom classification during a power loss.

a) The FFUs with Cleanroom Temperature & Humidity sensors located beneath shall be connected to Emergency Power. Therefore upon a loss of Normal Power, the T&H sensor would still be able to sense the Cleanroom conditions accurately, to keep the Cleanroom control function ongoing.

13.0 Instrumentation:

13.1 The following Instrumentation shall be installed and monitored/ controlled via FMCS. See **Table 12.**

Table 12: Instrumentation to be installed

Instrumentation	Type and Installation Location
Cleanroom Temperature and Humidity Transmitters	• Install 2 at each Temperature Control Zone (Quantity: 1 duty and 1 backup)
Cleanroom Differential Pressure Transmitter	• Install minimum 3 in FAB Cleanroom area
Dry Cooling Coil Leaving Air Temperature Transmitter	• Install at each Dry Coil Unit

13.2 The corresponding Cleanroom Dewpoint temperature shall also be shown on the SCADA screen for each Temperature and Humidity sensor.

14.0 Construction Stage Milestone

The following Construction Stage protocol shall be followed, in accordance with the Global Construction Cleanroom Protocol Standard.

a) **Stage 1 –** Rough Construction Stage - Building Shell Enclosed.
b) **Stage 2 –** Establishing the Clean Zone, Clean Zone Perimeter Boundaries Enclosed.
c) **Stage 3 -** Cleanroom Walls, Floors, and Ceiling Assembly Completed. One or more MAU units installed and operational in “Manual” mode, with filters installed including HEPA Filters, for early run operation to achieve start of Cleanroom Blowdown.
d) **Stage 4 –** Initial Cleanroom Certification; As-Built Status Achieved; Ceiling FFUs with HEPA Filters installed and operational.
e) **Stage 5 –** Intermediate and Final Qualification of Cleanroom and Installation of Production Equipment; Operational Status Achieved; Certification Testing.

15.0 Cleanroom Certification and Testing:

15.1 **Testing Requirements:** The following Cleanroom Certification Tests shall be performed during Cleanroom Certification:

a) Cleanroom Ceiling and Leakage Test
b) Cleanroom Classification via ISO 14644-1:2015 – Airborne Particle Test
c) Lighting Level test – minimum one point per 100m2
d) Sound Pressure Level – minimum one point per 100m2
e) Room Pressurization
f) Temperature and Humidity uniformity – minimum one point per 100m2, at a height of 1.0 to 1.2m above the Cleanroom Raised Floor
g) Floor resistance measurement – minimum one point per 100m2
h) Floor Vibration Testing
i) MAU HEPA Filter Leakage Test

15.2 **Cleanroom Construction Status Definitions**

15.2.1 **As-Built -** Condition where the cleanroom or clean zone is complete with all services connected and functioning but with no equipment, materials, or personnel present (ISO 14644-1:2015).

15.2.2 **At-Rest -** Condition where cleanroom or clean zone is complete with equipment installed and operating in a manner agreed upon, but with no personnel present (ISO 14644-1:2015).

15.2.3 **Operational -** Agreed condition where the cleanroom or clean zone is functioning in the specified manner, with equipment operating and with the specified number of personnel present (ISO 14644-1:2015).

15.3 **Initial Certification -** Completed once the "As-Built" status is achieved. Testing is to demonstrate that airborne and surface particle counts, temperature and humidity, surface resistance, and air balance and flow are within project specifications. Testing is also to confirm that the ceiling grid and HEPA filters are sealed and functioning as designed.

15.4 **Intermediate Certification -** Completed once the "At-Rest" status is achieved. Testing is to demonstrate that airborne and surface particle counts, and temperature and humidity are within project specifications.

15.5 **Final Certification -** Completed once the "Operational" status is achieved. Testing is to demonstrate that all environmental criteria are within project specifications.

16.0 Control Sequence Philosophy:

16.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 13**. Refer to the specific Sequence of Operations for each project for specific details.

Table 13: System Control Logic

Control Parameter	Control Logic
Cleanroom Drybulb Temperature Control	• Cleanroom Drybulb Temperature control shall be achieved via the Dry Cooling Coil modulating control valves. • Each control valve position shall modulate based on the Cleanroom Drybulb Temperature from the corresponding temperature control zone.
Cleanroom Humidity Control	• Cleanroom humidity control shall be achieved via Dewpoint Control of the MAUs. • The MAU discharge dewpoint shall be achieved via fixed Control or Auto-Cascade Control based on Cleanroom dewpoint
Cleanroom Pressurization Control	• Cleanroom Pressure Control shall be achieved via the Direct Pressure-Differential Control (DP) method. A pressure differential sensor measures the pressure difference between the Cleanroom and adjacent space (outside ambient, via the ZRP). • The Makeup Air Unit fan speeds shall be automatically modulated, to adjust the Makeup airflow amount supplied to the Cleanroom, to achieve the required pressure differential.

__**Note:**__ The control philosophy for the MAUs, Exhaust, and Chilled Water Systems will not be covered in this Standard.

16.2 The following table shows the control valve fail positions that shall be configured. See **Table 14.**

Table 14: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	**Loss of Electricity**	**Loss of Control Signal**
Dry Cooling Coil Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State

16.3 Any control valve actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

16.4 Upon a loss event of control air, power, or PLC signal, and then after the utility is restored, the control valve output command/position shall be set to the same command/position it was operating at prior to the loss event.

16.5 All control valves shall have a manual override bypass, to allow manual operation of the valve upon a failure.

16.6 SCADA Monitoring and Alarms: The following SCADA Alarms shall be enabled for Cleanroom Temperature and Humidity:

- HighHigh: 25°C and 60% RH
- High: 24°C and 57.5% RH
- Low: 20°C and 42.5% RH
- LowLow: 19°C and 40% RH

16.7 On the SCADA graphics screen for Cleanroom Temperature and Humidity, each T&H transmitter location shall also show the equivalent Dewpoint Temperature in SCADA.

17.0 Inter-Discipline Coordination Matrix:

17.1 The following table shows the provisions that shall be provided by each of the following disciplines and coordinated accordingly. See **Table 15.**

Table 15: Provisions Provided by Each Discipline

Discipline	Provision
Architectural	• Cut openings in walls and floors for piping penetrations, and seal openings after piping installation • Provide adequate fire stopping at all required fire wall penetrations There shall be no air infiltration or leakage from the outside spaces into the Cleanroom including the Return Air Chase or Interstitial Plenum. This includes from, but not limited to, the following measures: • Building construction seams, joints, and gaps shall be covered with separation panels and sealed to prevent infiltration and leakage. • All joints in walls, floors, and ceilings shall be of vaportight construction.

Discipline	Provision
	• Sealing and caulking with cleanroom approved sealant shall be used to seal all gaps between panels and gaps in cut-outs.
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the centralized equipment. Room temperature shall be designed to maintain the space temperature range of 24°C to 28°C. • Provide Makeup Air Unit(s), heating water and chilled water supply piping, supply ductwork, and accessories as outlined in this Standard • Provide Exhaust Fan(s), exhaust ductwork, and accessories as outlined in this Standard • Provide chilled water piping and accessories as outlined in this Standard • Provide floor drains located suitably to allow proper draining of equipment, condensate drains and auto-drains, piping, and air vents
Electrical	• Provide power wiring for all Electrical equipment/components • Provide grounding for all Electrical equipment/components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Water / Wastewater	• Provide drain source for the Condensate Drain water piping
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, dewpoint, flow, etc. • Integrate FFU System Controller with FMCS/SCADA

18.0 Quality Inspection and Testing:

18.1 **Inspection and Test Plan (ITP) –** Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

18.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping, Ductwork, and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning, Flushing, and Purging
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor’s Procedure to allow equipment to run in MANUAL mode. Also considered “Early Run” Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical

failure modes.

- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.2 **Cleanroom QA/QC Inspection** – Prior to testing, QA/QC inspection shall be performed on all installed equipment, piping, ductwork, and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards.

19.0 Startup Commissioning / Site Acceptance Test

19.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Commissioning / Site Acceptance Test.

19.1.1 Cleanroom Temperature

- Verify Dry Cooling Coil CHW Control Valve Function through the entire operating range (0 to 100%)
- Confirm Dry Cooling Coil CHW Control Valve control in response to Cleanroom Drybulb Temperature fluctuations

19.1.2 Cleanroom Humidity

- Confirm Makeup Air Unit discharge setpoint control in response to Cleanroom Dewpoint Temperature fluctuations

19.1.3 Cleanroom Pressurization

- Confirm Makeup Air Unit fan speed setpoint control in response to Cleanroom Pressure fluctuations

19.1.4 Cleanroom FFUs

- Confirm FFU fan speed setpoint and speed adjustment control through the entire operating range (0 to 100%)

19.1.5 Instrumentation and Controls

- Confirm Auto-switch to backup Cleanroom Temperature & Humidity transmitter for a single Temperature Control Zone, upon failure of main Cleanroom Temperature & Humidity transmitter in that zone
- Confirm Auto-switch to the other backup Cleanroom Pressure transmitters, upon failure of one Cleanroom Pressure transmitter
- Confirm all control valves fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA

20.0 Reference Specifications:

20.1 The following Micron Global Facilities Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 16.**

Table 16: Reference Construction Specifications

Spec Section	Spec Contents	Remarks
Section 13 21 13	Clean Rooms	
Section 09 58 13	Cleanroom Gasketed Ceiling System	
Section 09 69 60	Cleanroom Access Flooring	
Section 10 22 20	Cleanroom Partitions	
Section 10 57 53	Cleanroom Specialties	
Section 23 34 40	Cleanroom Fan Filter Units	
Section 21 10 00	Automatic Sprinkler System	
Section 23 77 03	Cleanroom Makeup Air Handling Units	
Section 23 41 00	Air Filters, HVAC	
Section 23 07 13	Ducting Insulation	
Section 23 07 19	HVAC Piping Insulation	
Section 23 05 29	Hangers and Supports for Piping	
Section 23 05 53	Mechanical Identification	

21.0 Document Control

21.1 **Approval**: GLOBAL_FAC_GFTT_DOC_APPR

21.2 **Notification**

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

21.3 **Final Viewer Access**

All Micron Team Members at all Sites

21.4 **Retention**

Adherence to the requirements specified in the
Global Facilities – Document Control Procedures

21.5 **Review**

Biennially (two years)

22.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	12/26/23